

Simultaneous Pairwise Reachability Filtering for Lifelong MAPF

Lekan Molu[†]

Abstract—Graph-based collision-free multi-robot path-finding plans are not immune to contacts/collisions when deployed on vehicles with continuous dynamics and disturbance. We study a steering filter for continuous execution of lifelong multi-agent path finding (MAPF) plans, where a three-dimensional Hamilton–Jacobi (HJ) backward-reachable tube supplies a numerical candidate barrier for ordered robot pairs and its gradients yield robust affine constraints on steering. The HJ filter intersects these constraints and selects the turn rate closest to the nominal command, with an empty intersection triggering a least-violation fallback. The filter is computed offline to serve all pairs, with constant interpolation cost per pair. Across 30 paired, 14-robot warehouse episodes, mean contacts fall from 52.47 under geometric execution to 1.07 under simultaneous filtering. The one-constraint ablation averages 0.73 contacts; the paired comparison does not establish an advantage for either barrier controller. On 25 solved continuous-time conflict-based search schedules, simultaneous filtering reduces mean contacts from 17.56 to 0.44, while increasing final goal distance from 0.98 to 7.50 cells. The results demonstrate contact reduction and its tracking cost.

I. INTRODUCTION

Multi-agent path finding (MAPF) consists in moving agents along *non-conflicting* paths in a sequence of $\{\text{move}, \text{wait}\}$ actions per agent across various unique start and goal states on the nodes of a graph, while avoiding collisions. Its applications traverse warehouse automation [1], robot routing [2], network routing [3], and aviation [4], *inter alia*.

(i) *cost-sums*, i.e., minimizing the cumulative arrival time steps that all agents take to arrive and settle at their goal; (ii) *makespan* i.e., maximizing the arrival time of all agents at their goal locations; or (iii) *fuel costs*, i.e., minimizing the total distance traveled whereupon only the *move* actions incur costs. Lifelong MAPF is the *rolling horizon* version of MAPF, whereupon an agent is assigned a new goal after reaching its previous goal location [5]: new plans are computed at every finite time step, then interleaved with execution [1].

To extend MAPF to physical phenomena, typically discrete MAPF plans are executed through continuous-time geometric models [6] or post-processing schedules. However, these abstractions do not directly represent turn-rate-limited execution under bounded disturbance. We thus ask whether pairwise relative configurations can be used to select collision-avoiding controls under bounded actuation and disturbance.

Hamilton-Jacobi (HJ) reachability [7–9] is a tool for evaluating the decidability of a dynamical system’s trajectories’ evolution throughout a state space. Decidable reachable systems are those where one can compute all states that can be reached from an initial condition in a *finite number of steps*. One can construct the so-called *backward reachable set/tube* (BRT) of a dynamical system so that adaptability of complex systems under uncertain dynamics can be certified for all constraints geometrically.

Thus, we pose agents’ pairwise interaction in relative coordinates [10] so that the game’s BRT collects relative configurations from which an adversarial vehicle can force contact within a finite horizon, against the protagonist agent’s best admissible steering. It provides a near-analytic characterization for turn-rate-limited vehicles [11], and the resulting set deforms with relative heading. Specifically, it expands into a long lobe for head-on configurations, where turn-rate limits restrict lateral clearance, and remains more compact for nearly parallel motion. More generally, a purely geometric collision predicate does not capture this heading-dependent avoidability geometry.

For states with vehicles $N > 2$, pairwise HJ collision avoidance with a mixed-integer control logic was established in Chen et al. [12] to assign pairwise avoidance responsibilities, providing a three-vehicle guarantee under stated initial-safety and assignment conditions, and evaluating larger fleets experimentally.

Grid-based joint-state reachability requires storage exponential in the number of state dimensions. We use a three-dimensional pairwise table and test whether one steering command satisfies all pairwise conditions,

$$\mathcal{U}_i = [-\bar{\omega}, \bar{\omega}] \cap \bigcap_{j \neq i} \{u_i : b_{ij}u_i + c_{ij} \geq 0\}, \quad (1)$$

where $i, j \in \{1, \dots, N\}$, $i \neq j$, identify an ordered pair, u_i is the local turn-rate control of agent i , and $\bar{\omega}$ is the maximum turn rate. Here b_{ij} and c_{ij} are the coefficient and offset obtained by expanding the robust pairwise barrier condition as an affine inequality in the ego turn rate u_i ; their construction is given in § IV. When (1) is nonempty, one agent can meet several avoidance requirements with one command. When it is empty, we report the incompatibility and do not solve the higher-level recovery problem. The continuous-time statement requires positive initial barriers, compliance with the model bounds, and satisfaction of every barrier inequality throughout execution. Our all-directed-pair constraints are conservative and do not imply a larger feasible set. *In this paper, we investigate whether one steering*

[†]The author is with the Amazon Industrial Robotics Group. However, the efforts reported in this article were executed outside of Amazon office/work hours, did not include using Amazon resources, datasets, or in-house expertise. Correspondence: lekanmolu@scriptedonachip.com.

command can satisfy all active pairwise barrier inequalities at once.

Our **contributions** are as follows. We

- formulate dynamics-aware continuous execution for lifelong MAPF as a runtime control problem over committed discrete paths and bounded unicycle dynamics;
- construct a pairwise relative-state reachability table and use it to derive robust affine steering constraints that can be evaluated in $O(1)$ per ordered pair, with total pair enumeration remaining $O(N^2)$;
- specialize simultaneous barrier filtering to scalar steering constraints derived from the pairwise table, using interval intersection and nominal-command projection with explicit infeasibility reporting;
- evaluate the filter on 30 paired lifelong warehouse instances and on execution of continuous-time conflict-based search (CCBS) schedules. We report contact counts, multi-threat constraint satisfaction, steering infeasibility, and goal-tracking error; a contact-count advantage over the one-constraint ablation remains inconclusive.

The rest of this paper is structured as follows: A non-exhaustive literature review is provided in § II; our formulation is detailed in § III and our approach is detailed in § IV. Results from numerical experiments are provided in § V and we conclude the paper in § VI.

II. RELATED WORKS

This section complements some of the related literature used in contextualizing our contribution in § I.

MAPF. Stern et al [13] cataloged the assumptions, conflicts, objectives and variants of MAPF necessary for establishing baselines. Beyond CBS and ECBS, Priority Inheritance with Backtracking (PIBT) is an adaptive, scalable and iterative, priority- and backtracking-based MAPF scheme that finds suboptimal solutions within a finite computation time [14]. Lazy constraints addition search for MAPF (LaCAM) [15] is a *complete* scheme that produces solutions for problems in a two-level hierarchical scheme — reporting when a solution exist or non-existence, otherwise. Empirically, it was demonstrated to work for reasonably large maps and with large multi-agents (up to 10,000 agents). Across this research inquiry line, the two conflict types are tied to occupancy: a vertex conflict when two agents hold a cell at one timestep, and an edge conflict when two agents swap along an edge.

Lifelong and warehouse-scale MAPF. Rolling-horizon collision resolution (RHCR) [1] decomposes the lifelong problem into windowed MAPF sequences, resolving conflicts within a bounded horizon w , and replanning as the window advances. The windowing allows re-allocations of goals to agents as windows advance and collision resolution is limited to the first h timesteps, $h \geq w$.

Conflict-based Search. A critical search tool for resolving MAPF on grids is *conflict-based search* or CBS [16]. CBS is a *complete* and *optimal* path-finding search method that works in a two-hierarchical search strategy: it resolves the MAPF problem optimally by keeping the spatio-temporal

position of an agent at each node at a higher level of a search *constraint tree* (CT); it then resolves the paths of an agent in the CT consistent with the constraints in a lower-level of the search hierarchy. Whenever two agents end up in a cell at the same time index after a plan, paths are replanned with the CT level adding new nodes and constraints.

Enhanced CBS (ECBS) keeps completeness of the solution at the cost of a resulting suboptimal algorithm at runtime. ECBS runs focal search at the CT and lower levels [17], while a parallel branch replaces scalar edge costs with a cost vector. Here, the *Pareto frontier* i.e., the set of all *undominated* solutions can grow exponentially in the graph size [18]. A**pex* bundles paths of approximately equal cost under a lower-bounding *apex*, merging bundles whenever a componentwise approximation factor admits it, so that it returns an approximate frontier in place of the full one [19]. Anytime A**pex* (A-A**pex*) is the anytime variant of A**pex*, working with multiple objectives to return solutions that altogether approximate the Pareto frontier better than the solutions found by other state-of-the-art multi-objective search algorithms [20].

Computing MAPF solutions for large multi-robot systems is NP-hard [21] with an exponential computational complexity in the agents' state space. State-of-the-art discrete-time and discrete-action algorithms only scale to dozens of agents *on structured instances* [15]. Our concern is the execution model: unit-time graph actions do not by themselves specify admissible steering under disturbance.

Continuous-time planning and execution. MAPF_R generalizes MAPF to real-space actions and costs [22] while CCBS [6] solves continuous-time instances by constraining unsafe action intervals and replanning paths. Temporal-plan-graph methods incorporate kinematic limits and execution dependencies [23, 24]; robust schedules address bounded delay [25]. Their assumptions differ from our adversarial steering model. While Chen et al. [12] choose binary avoidance responsibilities, our interval filter enforces every active directed inequality when feasible.

Reachability and safety filters. Reachability safety filters override nominal controls when needed to preserve a safe set [26]. Control barrier functions (CBFs) impose safety inequalities through online optimization [27]. Wang et al. [28] enforced multiple robot-collision constraints while minimizing deviation from nominal control. Choi et al. [29] connect HJ reachability to robust barrier-value functions and optimization-based control. In this paper, we specialize simultaneous constraint enforcement and reachability-based filtering to ordered-pair numerical tables and scalar steering constraints for disturbed execution of MAPF schedules, and measure the resulting contact-tracking tradeoff.

III. PROBLEM FORMULATION

Suppose that $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a graph and $\mathcal{A} = \{1, \dots, N\}$ denote agents on the graph nodes. At each replanning time, agent i has a current graph state, a goal, and a committed path $\pi_i : [0, T_p] \rightarrow \mathcal{V}$ over the planner's execution window T_p . The graph planner provides nominal paths that do

Discrete MAPF CBS/ECBS/RHCR Search

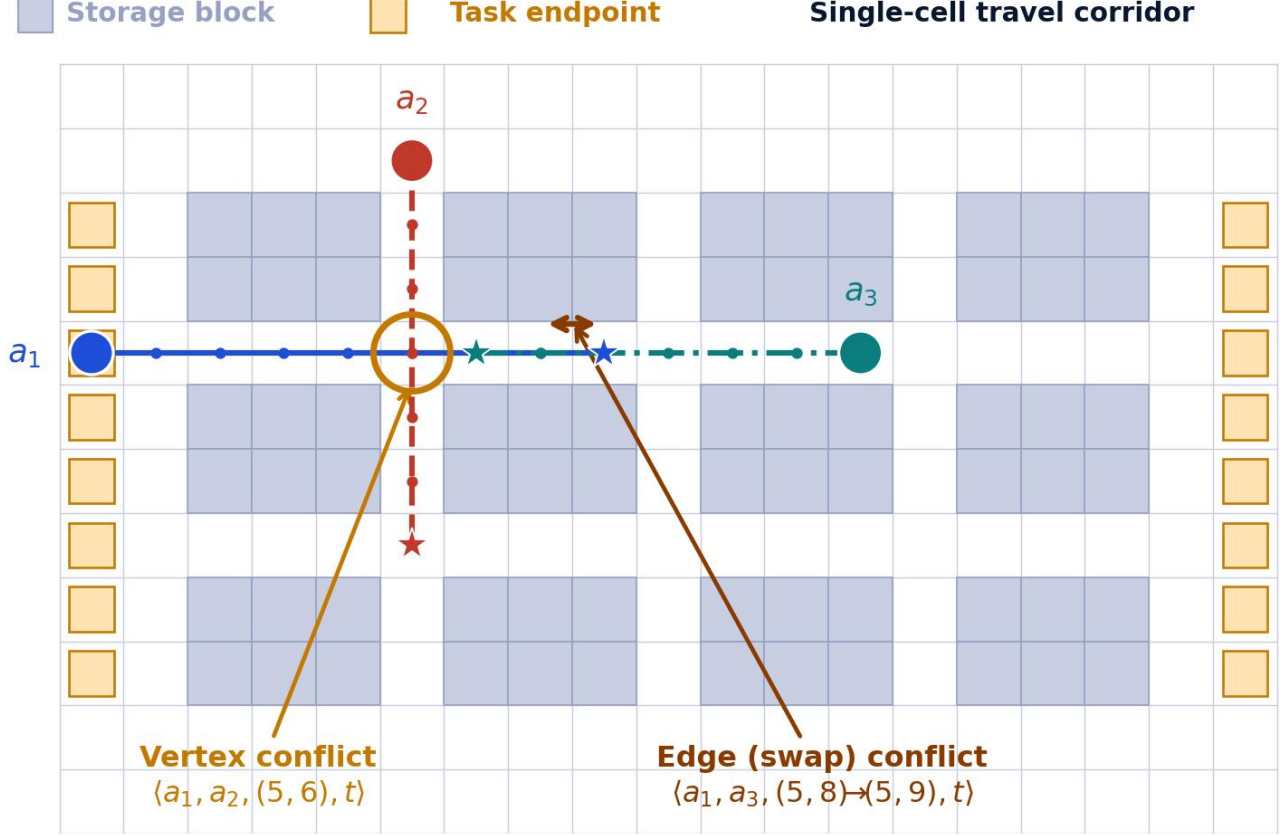


Fig. 1. Discrete-time and action abstraction of CBS, ECBS and RHCR search. Storage blocks are separated by single-cell travel corridors, and task endpoints are at border columns. A vertex conflict is marked at a corridor junction, and an edge conflict where two agents traverse one edge in opposition.

not necessarily represent the continuous vehicle dynamics employed during execution. The state of agent i is denoted $z_i = (p_i, \theta_i)$, with position $p_i \in \mathbb{R}^2$ and heading $\theta_i \in \mathbb{S}^1$. Its execution dynamics are

$$\dot{p}_i = v_i \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \end{bmatrix} + d_i, \quad \dot{\theta}_i = u_i, \quad (2)$$

with bounds $0 \leq v_i \leq \bar{v}$, $|u_i| \leq \bar{\omega}$, and $d_i \in [-D, D]^2$, where $D \geq 0$ is known. The path supplies nominal speed and steering commands and the filter modifies only the turn rate.

Agents are modeled as discs of radius r with continuous execution that is collision-free over an interval $[0, T]$ when

$$\|p_i(t) - p_j(t)\|_2 > 2r \quad \forall i \neq j, \quad t \in [0, T]. \quad (3)$$

Problem 1 (Dynamics-aware lifelong-MAPF execution):

Given a graph instance, committed paths $\{\pi_i\}_{i=1}^N$, current continuous poses, nominal controls, and a pairwise reachability table, compute bounded runtime controls that remain close to the nominal controls while maintaining all active pairwise safety conditions whenever a common admissible control exists.

For each monitored ordered pair (i, j) , each incident interpolation cell supplies one gradient branch, indexed by

$\ell \in \mathcal{L}_{ij}$. Each branch yields the robust affine constraint

$$b_{ij\ell}u_i + c_{ij\ell} \geq 0. \quad (4)$$

Let $\mathcal{M}_i$ contain the opponents within the table domain. The common admissible control set is

$$\mathcal{U}_i(z) = [-\bar{\omega}, \bar{\omega}] \cap \bigcap_{j \in \mathcal{M}_i} \bigcap_{\ell \in \mathcal{L}_{ij}} \{u_i : b_{ij\ell}u_i + c_{ij\ell} \geq 0\}. \quad (5)$$

When $\mathcal{U}_i(z)$ is nonempty, the runtime command is the projection

$$u_i^* = \arg \min_{u_i \in \mathcal{U}_i(z)} |u_i - u_i^{\text{nom}}|^2. \quad (6)$$

When $\mathcal{U}_i(z)$ is empty, the method reports an infeasibility event. The experimental least-violation fallback returns a bounded command that maximizes the minimum residual, but it is not a safety certificate.

Let $\tilde{V}$ be the trilinear table interpolant and set $h_{ij} = \tilde{V}(x_{\text{rel}}(i, j)) - \varepsilon_g - m$, with margin $m \geq 0$. For spatial grid spacings $\Delta x_1, \Delta x_2$, choose $\varepsilon_g = \frac{1}{2} \sqrt{\Delta x_1^2 + \Delta x_2^2}$. The table satisfies $V \leq g$ at every node, with $g = \sqrt{x_1^2 + x_2^2} - 2r$. Interpolating this distance function overestimates it by at most ε_g , so $h_{ij} \leq \|p_i - p_j\|_2 - 2r - m$. This geometric enclosure does not certify HJ approximation accuracy.

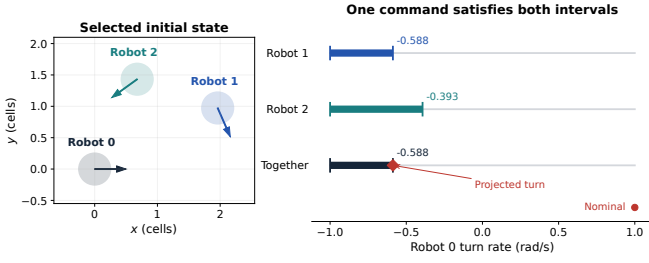


Fig. 2. A selected three-robot state illustrates the compatibility test. Robot 0 faces two simultaneous turn-rate restrictions; the feasible intersection is the interval selected by the filter. The drawing reports a mechanism example, not a statistical benchmark.

Proposition 1 (Conditional continuous-time safety):

Suppose that the geometric enclosure holds, every directed pair initially has $h_{ij} > 0$, and the executed trajectories are absolutely continuous and remain in the table domain. If $\dot{h}_{ij} + \kappa h_{ij} \geq 0$ almost everywhere for every pair, with $\kappa > 0$, then every pair remains separated by more than $2r$ at every finite time.

Proof. The interpolant is locally Lipschitz, so that each h_{ij} is absolutely continuous along the trajectory. Integrating $\frac{d}{dt}(e^{\kappa t} h_{ij}) \geq 0$ gives $h_{ij}(t) \geq e^{-\kappa t} h_{ij}(0) > 0$. The geometric enclosure then gives $\|p_i(t) - p_j(t)\|_2 > 2r$. Enforcing every incident gradient branch covers grid-cell interfaces.

Remark 1 (Scope of Prop. 1): Continued feasibility and enforcement between samples are assumptions of the proposition, not consequences of a finite-horizon table. Our sampled filter and least-violation fallback do not establish them; the experiments assess contact reduction rather than certified fleet safety.

IV. PAIRWISE BRT-INFORMED LIFELONG MAPF

Fig. 1 illustrates a typical 12×19 warehouse setup grid with storage block rows (grey-boxes) that are separated by single-cell travel corridors (contiguous transparent boxes). It possesses 156 free cells out of 228 total cells. Agent task endpoints are located on the left and right borders of the storage columns (shaded light yellow). Rows 5 and column 6 constitute the travel corridors. We see that vertex conflict occurs when agents $a1$ and $a2$ both demand using the corridor junction, located at cell (5, 6). Similarly, an edge swap conflict arises when both agents $a1$ and $a3$ traverse the same edge (row 5) along opposite directions. CCBS handles geometric conflicts through continuous-time unsafe intervals, shown in Fig. 2: one turn rate must satisfy both active pairwise constraints.

Tasks are lifelong: an agent that reaches its endpoint is immediately reassigned a new goal and the floor is setup such that agent density can seed adequate conflict patterns and frequency for our evaluations. 14 agents occupy 9.0% of free cells, against 4.7% and 11.0% for [5]’s 60- and 140-agent fulfillment runs.

A. Agent Dynamics

Plans are executed on unicycle dynamics with bounded turn rate and additive disturbance,

$$\dot{x} = v \cos \theta + d_x, \dot{y} = v \sin \theta + d_y, \dot{\theta} = \omega, |\omega| \leq \omega_{\max}. \quad (7)$$

Robots move at $v = 1$ cell/s, stopping within 0.05 cell of the current target, with $\omega_{\max} = 1$ rad/s. The nominal controller is $\omega = \text{clip}(4\Delta\theta, -1, 1)$ where each robot has radius 0.25 cell.

B. The pairwise game in relative coordinates

We set agent i as the defending ego vehicle for an ordered pair (i, j) and we let $R(\theta_i)$ be the rotation matrix that rotates body coordinates into the world frame. Define $(x_1, x_2)^\top = R(\theta_i)^\top(p_j - p_i)$ and $x_3 = \text{wrap}(\theta_j - \theta_i) \in [-\pi, \pi)$. The relative state is $x_{\text{rel}}(i, j) = (x_1, x_2, x_3)$. With speeds v_i, v_j and turn rates u_i, u_j , the undisturbed relative dynamics are

$$\begin{aligned} \dot{x}_1 &= v_j \cos x_3 - v_i + u_i x_2, \\ \dot{x}_2 &= v_j \sin x_3 - u_i x_1, \quad \dot{x}_3 = u_j - u_i. \end{aligned} \quad (8)$$

The contact set is $\{x_{\text{rel}} : g(x_{\text{rel}}) \leq 0\}$, with $g(x_{\text{rel}}) = \sqrt{x_1^2 + x_2^2} - 2r$ and contact distance $2r = 0.5$ cell.

C. Windowed value function and conflict set

We choose a reachability horizon of $w = 0.6$ s, independent of the planning window and replanning period. With look-back time $\tau \geq 0$, the defender-maximizing value satisfies

$$\partial_\tau V = \min\{0, H(x_{\text{rel}}, \nabla V)\}, \quad V(0, \cdot) = g(\cdot), \quad (9)$$

with $H = \max_{|u_i| \leq \bar{\omega}} \min_{|u_j| \leq \bar{\omega}, \|\eta\|_2 \leq 2\sqrt{2}D} \nabla V^\top [f + (\eta_1, \eta_2, 0)^\top]$. Here f is the undisturbed relative dynamics and η bounds the relative translational disturbance. The nonpositive update forms the numerical tube

$$\mathcal{L}(w) = \{x_{\text{rel}} : V(w, x_{\text{rel}}) \leq 0\}. \quad (10)$$

The offline game fixes $v_i = v_j = 1$ cell/s and encloses the rotated relative disturbance box in a ball of radius $2\sqrt{2}D$. Here, tube membership implies that the adversary can force contact within w despite admissible ego steering. The numerical table has no certified approximation-error bound. Runtime constraints use the commanded speeds, including stops; the fixed-speed game’s inevitability interpretation does not cover those stopped states. We use the fixed slice $V(w, \cdot)$ as a candidate barrier and do not infer continuing feasibility from it.

Definition 1 (Dynamics-aware conflict): Agents i and j conflict at time t if $x_{\text{rel}}(i, j) \in \mathcal{L}(w)$ or $x_{\text{rel}}(j, i) \in \mathcal{L}(w)$, evaluated at their current continuous poses.

Figure 3 plots the numerical table used by the controllers. Its zero-level set changes with heading, extending ahead of the ego vehicle in the head-on case and remaining more compact at parallel headings. About 2.8% of its grid nodes have nonpositive value at $w = 0.6$ s. This is a domain-dependent geometric statistic, not the observed frequency of intervention or a certified inevitability fraction.

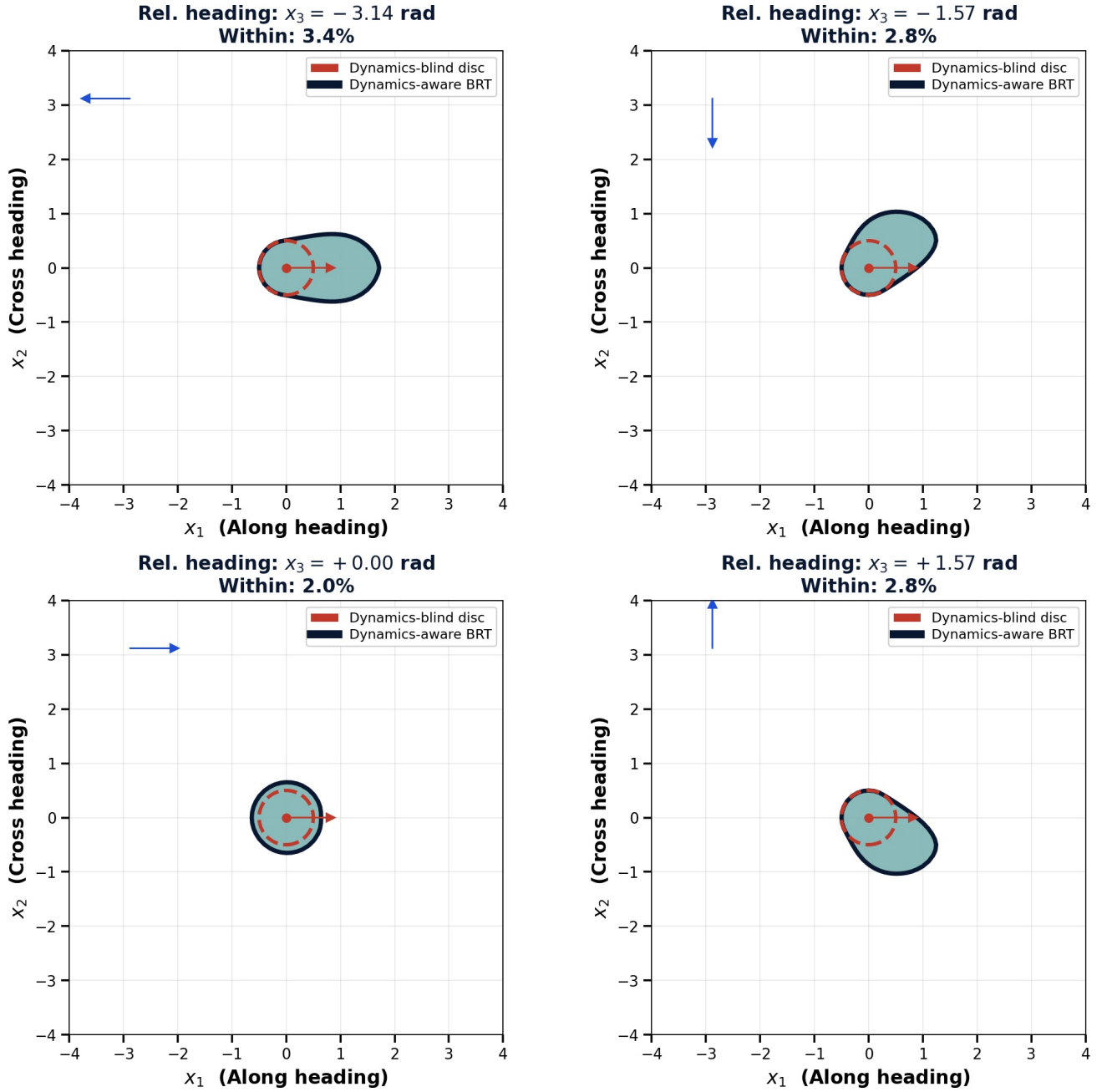


Fig. 3. The conflict set at four relative headings, with the dynamics-blind disc dashed for comparison. The red arrow shows the ego agent's heading, fixed along the x_1 axis; the blue arrow shows the other agent's heading, whose orientation is the relative heading x_3 .

D. Offline computation and runtime cost

A first-order local Lax–Friedrichs solver computes an $81 \times 81 \times 64$ table on $[-4, 4]^2 \times [-\pi, \pi)$. One offline solve serves the homogeneous fleet. Interpolation costs $O(1)$ per ordered pair, and all-pairs evaluation and feasible interval intersection cost $O(N^2)$ overall. The implemented least-violation fallback evaluates all pairwise row intersections: for K_i rows at robot i , it costs $O(K_i^3)$, giving an $O(N^4)$ dense-fleet worst case.

Numerical tube membership in the table lookup domain means $\tilde{V} \leq 0$ and the margin-adjusted barrier is nonpositive

when $h_{ij} \leq 0$. A row is active when $c_{ij\ell} - \bar{\omega}|b_{ij\ell}| < 0$, so that it excludes at least one bounded turn rate, possibly while $h_{ij} > 0$. Outside the table, the implementation omits a pair only if its separation exceeds $2r + 2(\bar{v} + \sqrt{2}D)\max(\Delta t, w)$; otherwise it reports unsupported coverage. This distance test excludes contact through the next sample under the speed and disturbance bounds.

E. Simultaneous runtime filter

At each control update, all robots use the same pose snapshot with the filter constructing monitored pairwise rows, intersecting admissible turn rates, and projects each nominal

Algorithm 1 Simultaneous pairwise runtime filter**Require:** Poses, speeds, nominal turns, table, model bounds

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1: for each agent  $i$  do
2:   check domain coverage; form all monitored rows
3:    $\mathcal{U}_i \leftarrow [-\bar{\omega}, \bar{\omega}]$ 
4:   for each pair and branch  $(j, \ell)$  do
5:      $\mathcal{U}_i \leftarrow \mathcal{U}_i \cap \{u_i : b_{ij\ell}u_i + c_{ij\ell} \geq 0\}$ 
6:   end for
7:   if  $\mathcal{U}_i \neq \emptyset$  then
8:      $u_i \leftarrow \text{proj}_{\mathcal{U}_i}(u_i^{\text{nom}})$ 
9:   else
10:    record infeasibility; apply least-violation turn
11:   end if
12: end for
13: Integrate the dynamics for one micro-step

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command into the resulting interval. An empty intersection triggers the least-violation fallback.

For a branch ℓ with gradient $\gamma = (\gamma_1, \gamma_2, \gamma_3)$, we minimize its directional derivative over $|u_j| \leq \bar{\omega}$ and independent $d_i, d_j \in [-D, D]^2$. Adding $\kappa h_{ij} - \rho$, with reserve $\rho \geq 0$, gives the affine lower bound $b_{ij\ell}u_i + c_{ij\ell}$, where

$$\begin{aligned}
b_{ij\ell} &= \gamma_1 x_2 - \gamma_2 x_1 - \gamma_3, \\
c_{ij\ell} &= \gamma_1(v_j \cos x_3 - v_i) + \gamma_2 v_j \sin x_3 \\
&\quad - \bar{\omega}|\gamma_3| - 2D\|R(\theta_i)\gamma_{1:2}\|_1 + \kappa h_{ij} - \rho, \quad (11)
\end{aligned}$$

where $\gamma_{1:2} = (\gamma_1, \gamma_2)^\top$. The disturbance term is the support function of the difference of two world-frame boxes. The experiment uses $\kappa = 2$, $\rho = 0$, $m = 0.1$, and actual commanded speeds, including zero near a target.

V. NUMERICAL RESULTS

We evaluate lifelong MAPF in the 12×19 warehouse shown in Fig. 1, using 14 robots and 30 seeds shared across controllers. At each replan stage, the planner computes a six-second path window and executes its first three one-second steps. Each episode comprises 40 cycles, totaling 120 seconds. Controllers share initial poses, tasks, priorities, and disturbance realizations within each seed. Forward Euler integration uses $\Delta t = 0.1$ s. Independent world-frame velocity errors follow $\mathcal{N}(0, 0.05^2)$, clipped to ± 0.12 cell/s per component. The $81 \times 81 \times 64$ BRT table uses a 0.6 s horizon and relative-disturbance bound $2\sqrt{2}D \approx 0.3394$ cell/s, with $D = 0.12$ cell/s. The runtime barrier margin is 0.1 cell.

Geometric execution tracks paths without avoidance control. *Matched flee* involves the robot turning away from an opponent while moving at 1 cell/s. It uses the same BRT table, barrier margin, execution model, and disturbance realizations as the one-constraint and simultaneous controllers. At each control sample, a robot selects the first higher-priority opponent in identifier order for which either directed margin-adjusted barrier is nonpositive. The one-constraint ablation projects the nominal turn rate onto the opponent's constraints with the smallest barrier, while the simultaneous

TABLE I
THIRTY PAIRED WAREHOUSE RUNS.

Controller	Contacts/run	Free runs	Multi-threat steps/run
Geometric	52.47	0/30	–
Matched flee	27.03	0/30	–
One constraint	0.73	19/30	523.40
Simultaneous	1.07	22/30	786.37

filter projects onto their common interval with an empty interval triggering the least-violation fallback.

A contact event starts when the swept Euler segments bring two centers within 0.5 cell. Initial and ongoing contacts are excluded; contact after separation counts as a new event. A successful multi-threat step requires at least two opponents to restrict a robot's turn rate, all evaluated constraints to hold, and all corresponding barriers to be nonnegative. The tracker commands 1 cell/s until it is within 0.05 cell of its current target, then commands zero speed; flee commands 1 cell/s.

Table I shows lower mean contact counts for both barrier controllers than for geometric execution or matched flee. The simultaneous controller averages 0.33 more contacts per episode than the one-constraint controller. However, the paired bootstrap 95% confidence interval for this difference, $[-0.70, 1.73]$, includes zero, so that the comparison does not establish a contact-count advantage for either controller. The simultaneous filter records 786.37 successful multi-threat agent-steps versus 523.40. Note that these are for different trajectories, so that the counts alone do not measure comparative success probability.

On average, no admissible turn rate satisfies all pairwise safety constraints at 8.56% of robot control updates (1438.50 of 16800 episode-wise). The filter then applies the least-violation fallback. These infeasibility events prevent us from invoking Proposition 1 to certify the full rollout.

CCBS execution on MAPF_R. Our CCBS implementation [6] solves 25 of 30 separate 14-robot warehouse instances within a 10-second limit. Each instance has distinct goals, cardinal edges, and robot-disc radius 0.25 cell. Independent continuous-time checks find no nominal path collisions; the minimum separation is $1/\sqrt{2}$ cell. Three execution controllers share each solved schedule, initial headings, and disturbance draws. The tracker follows the active segment's endpoint, commands zero speed during scheduled waits, and retains fractional segment times. Each simulation ends five seconds after the last scheduled arrival, averaging 27.59 s.

The MIP adaptation uses Chen et al.'s binary responsibility constraints [12]: each robot avoids at most one opponent, and a pair cannot receive reciprocal assignments. It uses our finite-horizon table, an activation threshold of 0.15 cell, and a cyclic priority extension to 14 robots. An assigned robot maximizes its opponent-specific worst-case value derivative.

Table II reports the separation of nominal planning from execution. The columns are delineated as mean contacts, contact-free runs, and mean final goal distance (cells). Across the 25 solved CCBS instances, the simultaneous filter re-

TABLE II
EXECUTION OF 25 CCBS SCHEDULES.

Execution	Contacts	Free runs	Goal error
Unfiltered	17.56	0/25	0.98
MIP adaptation	7.88	0/25	1.38
Simultaneous	0.44	20/25	7.50

duces contacts by an average of 17.12 per simulation relative to unfiltered execution and 7.44 relative to the MIP adaptation. The paired bootstrap 95% confidence intervals for these reductions are [15.32, 18.80] and [6.04, 8.84], respectively. At the end of the simulations, robots using the simultaneous filter remain an average of 7.50 cells from their goals, compared with 0.98 under unfiltered execution and 1.38 under the MIP adaptation. Thus, fewer robot–robot contacts come at the expense of goal tracking. Task completion follows planned grid positions rather than measured arrivals; we therefore report contact counts and final goal distance, but not task throughput.

VI. CONCLUSION

A collision-free MAPF plan does not ensure contact-free execution under bounded turning and disturbance. Our filter uses pairwise reachability gradients to test if one steering command can satisfy several avoidance constraints. It reduces contacts relative to geometric execution and *matched flee* in the evaluated warehouse, but does not establish a contact-count advantage over the one-constraint ablation. On CCBS schedules, lower contact counts accompany larger final goal distances. These results support the use of reachability constraints during execution while exposing the tracking cost of avoidance filtering.

The pairwise table is reusable, however fleet-wide evaluation still enumerates pairs. A complete navigation system requires obstacle constraints, recovery after infeasibility, and task completion based on physical arrivals.

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